OWRT-DETR: A Novel Real-Time Transformer Network for Small-Object Detection in Open-Water Search and Rescue From UAV Aerial Imagery

Shuai Ma, Yihong Zhang, Ling Peng, Chen Sun, Baolong Ding, and Yongdong Zhu

***Abstract*— Uncrewed aerial vehicle (UAV) object detection is crucial in open-water search and rescue missions. Due to varying perspectives and altitudes of UAV images, the apparent size of objects varies significantly. Challenges such as insufficient feature representation and background confusion make open-water object detection particularly difficult. Currently, deep learning-based detection methods rely on convolution to extract features at a fixed spatial scale. This limited receptive field leads to insufficient feature representation, causing false detections and missed detections, which severely impact detection accuracy. This article proposes an efficient feature-enhanced real-time detection network based on Transformer architecture, called OWRT-detection transformer (DETR), to address the challenges of diverse UAV image detection in open-water. To the best of authors' knowledge, a Transformer-based detection network has not yet been explored for open-water UAV images. OWRT-DETR incorporates a cross-scale feature pyramid interaction module (CFPIM), multiscale sensing fusion (MSSF), and small-object enhancement module (SOEM). These modules enhance cross-scale interaction, cross-channel spatial global association, and local perception of the network, while avoiding increased complexity, improving weak feature representation of small targets, and suppressing easily confused backgrounds. Three public datasets are used to validate the effectiveness of OWRT-DETR. OWRT-DETR achieves an average precision (AP) of 51.5%, 45.9%, and 50.6% on the SeaDronesSee, aerial dataset of floating objects (AFOs), and aerialbus ship (AS) datasets, exceeding the performance of several state-of-the-art models. To ensure efficiency and reduce computational resources, OWRT-DETR is optimized by reconstructing the backbone network using partial convolution (PConv) and Rep, resulting in Light-OWRT-DETR. Compared with OWRT-DETR, Light-OWRT-DETR is faster, uses fewer parameters, requires less computational power, and achieves higher accuracy. The code will be available at https://github.com/mshauima/OWRT-DETR**

***Index Terms*— Detection transformer (DETR), multiscale fusion, open-water search and rescue, small-object detection, uncrewed aerial vehicle (UAV) aerial imagery.**

Received 14 November 2024; revised 20 March 2025; accepted 10 April 2025. Date of publication 15 April 2025; date of current version 6 May 2025. This work was supported by Shanghai Industrial Collaborative Innovation Project Foundation under Grant XTCX-KJ-2023-2-18. *(Corresponding author: Yihong Zhang.)*

The authors are with the College of Information Science and Technology, Donghua University, Shanghai 200051, China (e-mail: 2232075@mail.dhu.edu.cn; zhangyh@dhu.edu.cn; 2222144@mail.dhu.edu.cn; 2232203@mail.dhu.edu.cn; 2232225@mail.dhu.edu.cn; 2232072@mail.dhu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3560928

I. Introduction

In RECENT years, with economic development, nearly 80% of transportation is completed by sea, and maritime transportation has become the main mode of cargo transportation. However, due to the changeable marine environment, maritime accidents caused by natural disasters such as strong winds, huge waves, storms, and tsunamis often occur [1], [2], [3], [4]. Maritime search and rescue serves as the final safeguard for ensuring the safety of human lives. The critical aspect of search and rescue operations is the precise determination of the relative positions between the distressed individual and the rescuer in open-waters. Currently, traditional search and rescue methods still largely depend on human involvement [5], [6], [7].

However, with the vast potential brought by the ongoing advancements in optical remote sensing, uncrewed aerial vehicles (UAVs) have emerged as a crucial tool for target detection, thanks to their high maneuverability, flexible viewing angles, low cost, and ease of deployment. The primary focus of UAV detection is to identify and locate objects in aerial images or videos captured by UAVs [8], [9]. Additionally, the pixel size of UAV imaging presents a significant challenge to detectors in classifying and locating small-objects, including targets smaller than 32×32 pixels, with dim features, low contrast, and insufficient information, especially in open-water environments [10], [11]. Traditional detection algorithms, such as support vector machine (SVM) [12] and histogram of oriented gradients (HOGs) [13], have demonstrated some effectiveness with low-resolution images, but remain inadequate in terms of real-time performance and accuracy [14], [15].

Over the past few years, deep learning technology has advanced significantly, and convolutional neural network (CNN)-based detection methods have rapidly emerged, greatly enhancing the performance of UAV target detection. With the ongoing advancements in deep learning technology, neural network-based detection algorithms have rapidly emerged, significantly improving the performance of UAV target detection. Deep learning models such as YOLO [16], Faster R-CNN [17], SSD [18], and others have demonstrated strong performance on commonly used object detection datasets such as COCO [19] and PASCAL VOC [20], and are widely applied in UAV image processing [21], [22]. Although the YOLO series

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:52:36 UTC from IEEE Xplore. Restrictions apply.

detectors have shown impressive real-time performance, they rely on convolution operations, possess limited receptive fields, and struggle to capture relationships between targets. However, their accuracy remains inadequate when faced with challenges such as complex backgrounds at sea, changes in lighting and weather, occlusion, target size variations, and the presence of multiple targets. To address these challenges, researchers are actively exploring ways to optimize detection models for UAV-based detection. Recently, the RT-detection transformer (DETR) [23] detector, which is based on the Transformer architecture, applies the self-attention mechanism only to high-level, semantically rich feature maps and integrates it with convolution operations to perform multiscale feature fusion. This improves the model's global modeling capability and effectively reduces computational redundancy. This design enhances detection speed while maintaining high accuracy. This study aims to design a high-precision small target detector to improve the detection performance of aerial images captured by UAVs operating over water, with the goal of applying it to future open-water search and rescue missions and addressing the need for extensive experiments and evaluations. Our proposed method addresses the issues of low detection accuracy and false positives (FPs) caused by the complex water surface background. Additionally, it is more effective and faster than state-of-the-art models. Moreover, to facilitate deployment, we proposed a simplified version of OWRT-DETR. To the best of authors' knowledge, we are the first to propose the use of Transformer architecture models for open-water detection, offering a novel option for future open-water UAV aerial image detection systems. The main contributions of this article are listed as follows.

1) A new variant of DETR, named OWRT-DETR, along with its lightweight version, L-OWRT-DETR, is proposed for open-water object detection tasks.
2) Three innovative and lightweight modules are proposed: small-object enhancement module (SOEM), cross-scale feature pyramid interaction module (CFPIM), and multiscale sensing fusion (MSSF) module. These modules are designed to address distinct challenges: SOEM enhances local perceptual capabilities for fine-grained details, CFPIM strengthens multiscale feature fusion across hierarchical representations, and MSSF establishes inter-scale correlations between local and global contexts. Collectively, they improve the weak feature representation of small-objects in UAV-based open-water detection tasks while effectively mitigating interference from complex backgrounds across different scales in open-water environments, thereby improving the model's robustness.
3) OWRT-DETR was evaluated through extensive experiments on three public open-water UAV remote sensing datasets. These experiments demonstrated the model's effectiveness. The model achieved significant improvements in high-precision performance for small targets in open-water, meeting real-time requirements and showing potential for future airborne applications.

II. Related Works

A. Object Detection in Open-Water

Open-water target detection is of great significance in search and rescue, maritime monitoring, environmental protection, and other fields. With the ongoing advancements in computer vision and deep learning, deep learning models have made significant strides in open-water applications. General detectors can be categorized into two types: detectors based on CNN architectures [16], [17], [18] and detectors based on Transformer [24] architectures [25], [26], [27], [28], [29].

CNN-based object detectors are widely applied in open-water due to their ability to rapidly extract local features through convolution operations and achieve high detection speed. FE-YOLO [30] employs variable convolution in the YOLO neck to fuse high and low features, eliminating the impact of top–down connections on object semantic information. CA-YOLO [31] and YOLOv7-sea [32] incorporate coordinate attention and parameter-free attention modules to identify the attention areas of small-objects at sea and enhance detection capabilities. YOLO-OW [33] effectively suppresses false detections and improves detection performance through a dual-path aggregation network and a detection filtering module.

Although the above research has improved the performance of CNN-based detectors, because of the limited receptive field in CNN, it is difficult to effectively capture global information, which leads to limitations when detecting complex backgrounds and open-water environments. By contrast, Transformer-based detectors can model global contextual information through the self-attention mechanism, process complex features, and long-distance dependencies, making them more suitable for detecting complex scenes. Therefore, the Transformer architecture detector excels in complex environments and intensive tasks, offers strong global modeling capabilities, flexible feature representation, and multitarget recognition, does not require anchor frames or postprocessing, and offers good scalability [34], [35], [36], [37].

B. Object Detection in Small-Object

Object detection models based on deep learning rely on obtaining high-dimensional features from the backbone network. However, in UAV aerial images, the extracted features of an object may only occupy one pixel on the output feature map. It is essential to combine multiscale features to fully represent the object. Lin et al. [38] introduced a feature pyramid to fuse low-level features with high-level features. Subsequently, models such as PANet [39], NAS-feature pyramid networks (FPN) [40], BiFPN [41], and ASFF [42] have been proposed for detection tasks, achieving notable results. In addition, models based on the Transformer architecture, Swin Transformer [43] employs a sliding window self-attention mechanism to fuse local and global features, effectively enhancing the ability to detect small targets. Deformable-DETR [26] uses deformable attention and multiscale feature fusion to better capture the details of small targets, thereby improving detection accuracy. Efficient-DETR [35] optimizes the attention calculation and feature extraction

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:52:36 UTC from IEEE Xplore. Restrictions apply.

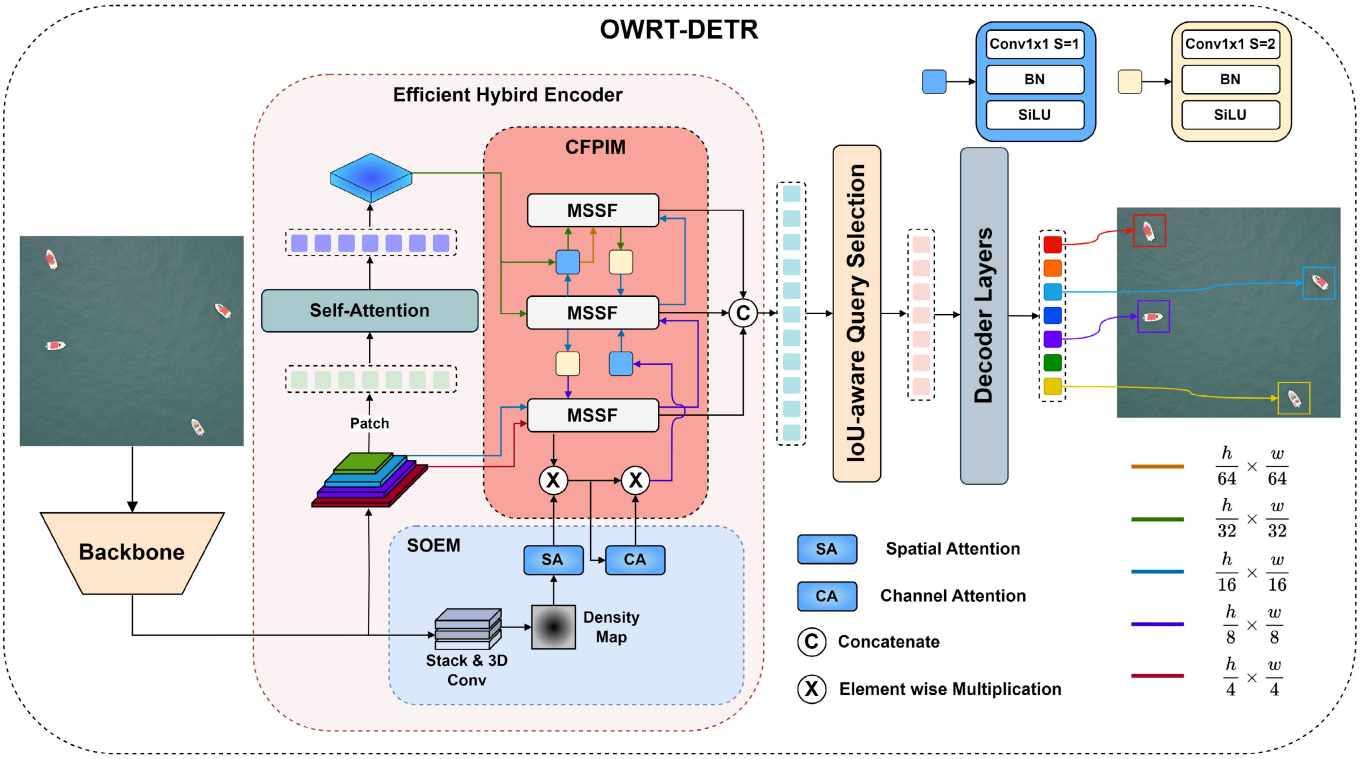


Fig. 1. Overall structure of OWRT-DETR model. The model consists of three main components: the backbone, encoder, and decoder. After passing through the backbone, the multiscale features are split into two paths, with one path directly fused with the cross-feature pyramid. The other path passes through the SOEM before entering the cross-feature pyramid fusion. Additionally, we reconfigure the fusion path and modify the fusion method. The decoder is then initialized with the initial object query via IoU-aware query selection. Finally, the decoder in the auxiliary prediction head iteratively optimizes the generated bounding box and a confidence score for the object query.

process, allowing it to focus quickly on key features when processing small targets, greatly improving efficiency. DINO-DETR [34] proposes a bidirectional interactive attention and multiscale enhancement module to improve the performance of small target detection. These methods all show that multiscale features can improve the performance of small target detection to some extent. In addition, feature extraction also affects the semantic information representation of the network. Enhancing the detailed information of the feature map before fusion can effectively improve the network's small-object detection performance. Wang et al. [44] and Cheng et al. [45] used spatial attention for object detection and multiscale feature refinement in UAV images.

Therefore, we designed a novel cross-scale interaction module that better combines low-level detailed information with high-level context information, alleviates the problem of feature loss, and enriches the model's contextual perception capabilities. Additionally, we stack feature maps from different layers, apply 3-D convolution to generate dense feature maps, and extract detailed information from these maps through spatial and channel attentions, enhancing the weak feature representation of small-objects and suppressing complex and cluttered backgrounds. This approach effectively improves the detection performance of small-objects.

III. Proposed Method

A. Overview

The proposed network is shown in Fig. 1 and primarily consists of four key components: the feature extraction network, the SOEM, the feature encoder, the feature decoder, and the prediction head. First, given an input image, multiscale features are extracted through the feature extraction network. Then, inspired by RT-DETR [23], only the S5 layer of the backbone network is used as input to the Transformer encoder. The S5 layer contains rich semantic features and maximizes the analytical capabilities of the Transformer encoder when processing large-scale serialized image data. Additionally, an SOEM is designed to generate a small-object attention feature map. This module (SOEM) significantly enhances the detection performance for small-objects, especially in the presence of complex background interference, frequently observed in UAV images. The efficient hybrid encoder employs a structure similar to FPN, facilitating intrascale feature interaction between the Transformer encoder, multiscale features, and small-object attention feature maps across scales, producing a visual feature sequence. This design improves robustness to scale variations in UAV images. Subsequently, a fixed number of image features are selected as the initial query for the encoder through an intersection over union (IoU)-aware query mechanism. Finally, the matching head optimizes the object query to achieve a one-to-one correspondence between the predicted bounding box and the ground truth.

B. Cross-Scale Feature Pyramid Interaction Module

This section introduces the CFPIM. CFPIM constructs an interactive module similar to the traditional pyramid network [40], introduces a novel fusion framework, and incorporates key SOEMs, designed to address the real-time detection

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:52:36 UTC from IEEE Xplore. Restrictions apply.

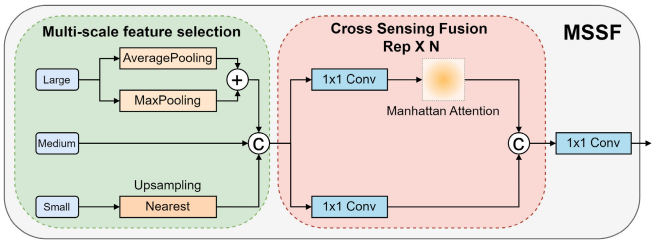


Fig. 2. Structure of MSSF.

challenges of small-objects in aerial drone images. The core component comprises two main elements: intrascale feature interaction based on self-attention and cross-scale feature fusion combining CNN and attention mechanisms. First, to reduce the computational redundancy caused by intrascale feature interaction in multiscale DETR, the interaction process is optimized. Intrascale feature interaction is limited to the S5 layer, which contains rich semantic information. This advanced feature layer can capture relationships among conceptual entities across objects, including their semantic relationships. In contrast, while the shallow layer of the network contains less semantic information, it retains more object edges, textures, and fine-grained details. These high-resolution features are crucial for detecting small targets, particularly in open-water scenes. Furthermore, as features at different levels exhibit significant differences in semantic expression and detailed retention, we constructed a multiscale cross-perception fusion mechanism that dynamically fuses semantic and detailed information from various feature levels. This mechanism ensures that the model captures accurate semantic information in complex scenes while minimizing the loss of critical details. The specific details of the MSSF are explained in Section III-C.

C. Multiscale Sensing Fusion

This section introduces the multiscale cross-perception fusion mechanism, which fuses features across different scales to generate a spatial scale feature map that contains both detailed and global semantic information. The structure of the module is illustrated in Fig. 2.

1) Multiscale Feature Selection: As a fundamental component of CNNs, the convolution block plays a crucial role in determining the accuracy and speed of detection models. Previous studies have demonstrated that multibranch convolutional paths [46], [47], [48], [49] often outperform single-branch paths [50], [51]. The original RT-DETR detector employs feature fusion at two scales in adjacent stages. However, the top-level feature map does not incorporate low-level feature fusion. Low-level feature maps contain a wealth of complex features, which are particularly important for small-object detection. Therefore, we select three scales for input; for larger feature maps, we use global average pooling and maximum pooling to combine them and obtain a reduced-size image, where maximum pooling highlights more prominent features, and average pooling captures smoother, overall features. By combining the two branches, we effectively capture the details of the large-scale feature map. For medium-sized feature maps, we apply a simple 1×1 convolution to preserve their dimensions. For small-sized feature maps, which contain richer semantic information, we employ "nearest" interpolation to match their spatial size with that of the medium-sized maps, while preserving discrete features. Finally, the three processed feature scales are concatenated and defined as follows:

$$X_{\text{out}} = (\text{GMP}(X_L) + \text{GAP}(X_L)) + \text{Conv}_{1\times1}(X_M) + \text{Interpolate}_{\text{nearest}}(X_S) \tag{1}$$

where X_L, X_M, and X_S represent the large-, medium-, and small-size feature maps of the input, respectively, and GMP and GAP are the global maximum pooling and global average pooling, respectively.

2) Cross-Sensing Fusion: Inspired by RMTNet [52], we introduced the Manhattan Self-Attention (MASA) mechanism into the fusion module. Temporal information is incorporated into the spatial dimension to establish a multiscale global modeling relationship. This module consists of two branches: one global branch using the MASA attention mechanism and one local branch that enhances local context information through $\text{Conv}_{1\times1}$. This architectural design provides a more comprehensive global receptive field when extracting image features while retaining local context information. Moreover, due to the MASA attention mechanism's inherent advantage in decoupling global attention, the fusion module incurs minimal computational overhead. RetNet [53] first proposed the retention mechanism for sequence modeling, introducing time decay into the language model. However, the field of image processing lacks a comparable time decay mechanism. Retentive networks meet vision transformers (RMT) introduced a spatial decay matrix, based on the retention mechanism, which incorporated temporal prior knowledge from the language model into visual tasks. The process of cyclic sequence modeling is as follows:

$$o_n = \sum_{m=1}^{n} \gamma^{n-m} \left(Q_n e^{in\theta}\right) \left(K_m e^{im\theta}\right)^{\dagger} v_m. \tag{2}$$

For the parallel training process, the above formula can be expressed as follows:

$$Q = (XW_Q) \odot \Theta, \quad K = (XW_K) \odot \overline{\Theta} \qquad V = XW_V \tag{3}$$

$$\Theta_n = e^{in\theta}, \quad D_{nm} = \begin{cases} \gamma^{n-m}, & n \geq m \\ 0, & n < m \end{cases} \tag{4}$$

$$\text{Retention}(X) = \left(QK^{\text{T}} \odot D_{nm}\right) V \tag{5}$$

where $\overline{\Theta}$ is the conjugate of Θ, and $D \in \mathbb{R}^{|x|\times|x|}$ includes the causal mask and attenuation, representing the relative distance in 1-D space. By introducing temporal prior knowledge into visual tasks, the spatial relationship matrix is remodeled as follows:

$$D_{nm}^{\text{Bi2d}} = \gamma^{|x_n - x_m| + |y_n - y_m|} \tag{6}$$

$$\text{BiRetention}(X) = \left(QK^{\text{T}} \odot D_{nm}^{\text{Bi2d}}\right) V \tag{7}$$

where $||$ represents the bidirectional modeling, and (x_n, y_n) represents the 2-D coordinates of each token in the images.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:52:36 UTC from IEEE Xplore. Restrictions apply.

The softmax function is applied to establish MASA attention. The attention mechanism is decomposed along the horizontal and vertical axes. The decomposed attention is described as follows:

$$\text{Attention}_i = \text{Softmax}\left(Q_i K_i^{\text{T}}\right) \odot D^i \tag{8}$$

$$\text{MaSA}(X) = \text{Attention}_H(\text{Attention}_W V)^{\text{T}} \tag{9}$$

$$X_{\text{out}} = \text{MaSA}(X) + \text{Local}(X) \tag{10}$$

where Local is $\text{Conv}_{1\times1}$, an enhancement module designed to improve the local receptive field. The input feature map is sequentially processed by the MASA attention module and the local module, after which the outputs are concatenated. This process captures both global and local relationships, introduces spatial prior knowledge, and effectively reduces computational complexity.

D. Small-Object Enhancement Module

Previous research has indicated that objects of varying sizes require different levels of semantic information [54], [55]. In remote sensing or drone imagery, detecting boats or individuals in water requires rich contextual information for the model to accurately learn the target's feature representation. To enhance the feature representation of small targets and fuse feature information across different channels, we designed an SOEM, which integrates small target information and multiscale features when fused with the S3 layer. This module consists of a dynamic multiscale fusion module and a spatial–channel attention guidance mechanism. By stacking the multiscale feature maps output by the Backbone, applying 3-D convolution to extract scale sequence features, generating small-object-enhanced feature maps, and producing attention maps through spatial attention, the attention maps are multiplied by the cross-scale interaction feature maps to enhance channel relationships. The spatial- and channel-attention feature map focuses on important regions, specifically foreground information, significantly improving small target detection. The calculation process is outlined in Algorithm 1. The definitions and interpretations of the symbols used in Algorithm 1 are provided, where $\mathbf{S_i}$ is the number of preprocessed feature layers, SPD_{conv} denotes the space-to-depth convolution operation, $\text{Interpolate}_{\text{nearest}}$ refers to the neighborhood interpolation upsampling method, Stack denotes the concatenation of feature maps along the channel dimension, Split refers to the partitioning of the feature map along the height and width dimensions, σ denotes the application of the sigmoid function for generating the attention map, X_{MSSF} represents the output feature map produced by the MSSF mechanism, and $\odot$ denotes the multiplication of elements.

Algorithm 1 SOEM

Preprocessing Image

Input:

$\mathbf{S_{i=\{2,4,5\}}}$, $\mathbf{S_2} \in \mathbb{R}^{\frac{h}{4}\times\frac{w}{4}\times4c}$, $\mathbf{S_4} \in \mathbb{R}^{\frac{h}{16}\times\frac{w}{16}\times16c}$,
$\mathbf{S_5} \in \mathbb{R}^{\frac{h}{32}\times\frac{w}{32}\times32c}$
$\mathbf{S_2'} \leftarrow SPD_{conv}(\mathbf{S_2})$, $\mathbf{S_2'} \in \mathbb{R}^{\frac{h}{8}\times\frac{w}{8}\times8c}$
$\mathbf{S_5'} \leftarrow Interpolate_{\text{nearest}}(\mathbf{S_5})$, $\mathbf{S_5'} \in \mathbb{R}^{\frac{h}{8}\times\frac{w}{8}\times8c}$
$\mathbf{X}_{Featuremap} \leftarrow Conv_{3D}\{Stack(S_2', S_4, S_5')\}$,
$\mathbf{X}_{Featuremap} \in \mathbb{R}^{\frac{h}{8}\times\frac{w}{8}\times8c}$

Enhancement:

$\mathbf{X}_h \leftarrow AvgP_h(\mathbf{X}_{Featuremap})$, $\mathbf{X}_h \in \mathbb{R}^{\frac{h}{8}\times\frac{w}{8}\times8c}$
$X_w \leftarrow AvgP_w(\mathbf{X}_{Featuremap})$, $\mathbf{X}_w \in \mathbb{R}^{\frac{h}{8}\times\frac{w}{8}\times8c}$
$\mathbf{X}_{Desitymap} \leftarrow Conv_{1\times1}(\mathbf{Concat}(X_h, X_w))$
$\mathbf{X}_{Desitymap} \in \mathbb{R}^{\frac{h}{8}\times\frac{w}{8}\times8c}$
$\mathbf{S}_h, \mathbf{S}_w \leftarrow \mathbf{Split}(X_{Desitymap})$, $\mathbf{S}_h, \mathbf{S}_w \in \mathbb{R}^{\frac{h}{8}\times\frac{w}{8}\times8c}$
$\mathbf{Attention}_h \leftarrow \sigma(\mathbf{S}_h)$, $\mathbf{Attention}_w \leftarrow \sigma(\mathbf{S}_w)$
$\mathbf{X}' \leftarrow X_{MSSF} \odot A_h \odot A_w$, $\mathbf{X}' \in \mathbb{R}^{\frac{h}{8}\times\frac{w}{8}\times16c}$
$\mathbf{Attention}_{Channel} = \sigma\left(Conv_{1\times1}\left(AvgP\left(\mathbf{X}'\right)\right)\right)$
$\mathbf{Attention}_{Channel} \in \mathbb{R}^{\frac{h}{8}\times\frac{w}{8}\times16c}$
$\mathbf{X}_{EnhancementMap} = \mathbf{X}' \odot \mathbf{Attention}_{Channel}$

E. LFE Backbone Network

The backbone of the DETR [25] model typically employs ResNet [56] as the primary feature extraction network. As a well-established deep learning network, ResNet addresses issues like gradient vanishing and explosion that occur in deeper layers of the network. However, the number of parameters in ResNet, composed of basic convolutional blocks within the DETR model (e.g., ResNet18), reaches 11.1712M, accounting for more than half of the entire network, introducing considerable computational complexity. Therefore, we aim to reduce the complexity of the backbone to meet real-time performance requirements. Previous research has explored model acceleration using approaches such as ShuffleNet [57] and GhostNet [58]. However, these approaches primarily reduce floating-point operations (FLOPs) to accelerate the model, which increases memory access. While FLOPs are reduced, the inference and runtime speeds remain slow, and the model's complexity is not fundamentally reduced. We restructured the backbone network inspired by FasterNet [59]; this network primarily adopts a new convolution structure, partial convolution (PConv), on which FasterNet is based. The working principle of PConv is illustrated in Fig. 3. During feature extraction, it applies ordinary convolution to selected channels, while keeping the remaining channels unchanged. This ensures efficient feature extraction while reducing computational complexity. The input feature map is $X \in \mathbb{R}^{h\times w\times c}$ with a convolution kernel size of $W \in \mathbb{R}^{k\times k\times c}$. For c_0 convolution kernels, the computed FLOPs is $h \times w \times k_2 \times c_2$. Since PConv only performs convolutions on selected channels, the size of the convolution kernel becomes $\overline{W} \in \mathbb{R}^{k\times k\times c_p}$ of the original, where c_p represents a fraction of c, with a partial ratio of $r = (c_p/c)$. If $r = (1/4)$, the FLOPs will be reduced to $(1/16)$ of the original. Research has demonstrated that multibranch convolution blocks typically outperform single-path convolution blocks. However, multibranch convolution blocks often result in reduced parallelism and increased inference latency. To balance inference speed and feature extraction, we draw inspiration from RepVGG [60] and use reparam convolution (RepConv) as the base layer. Building on ResNet, we constructed a new block. Lightweight and fast extraction block (LFEB) consists of multiple branches during training.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:52:36 UTC from IEEE Xplore. Restrictions apply.

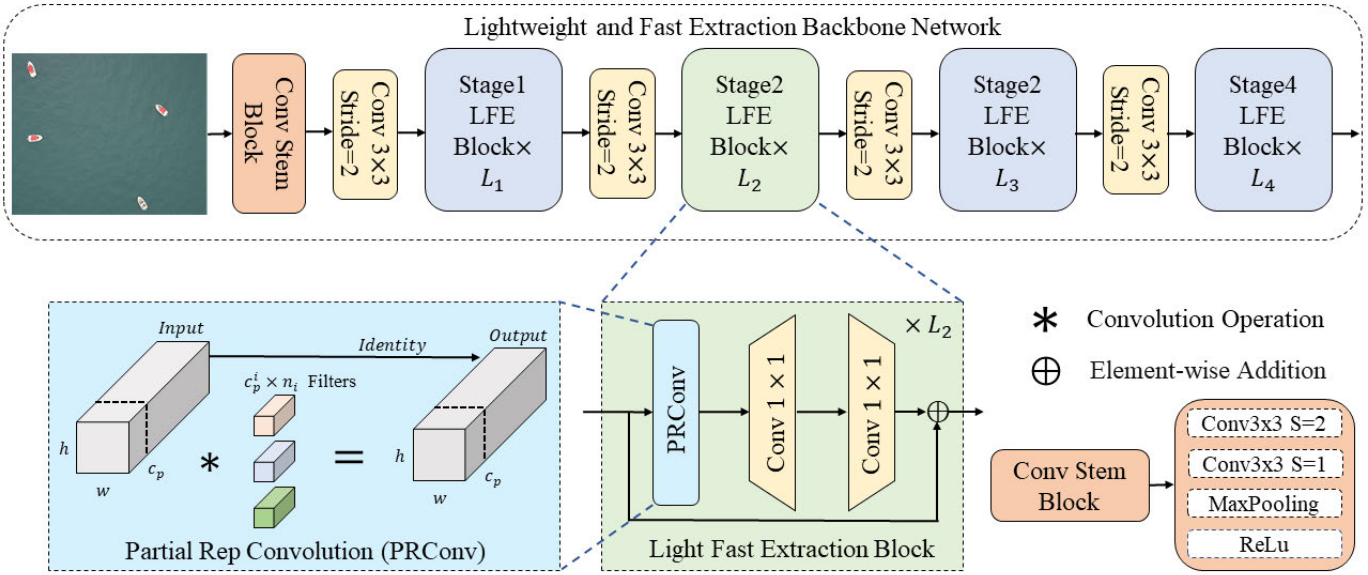


Fig. 3. Backbone structure of L-OWRT-DETR. The Backbone architecture is organized into five stages. Each stage comprises multiple Light Fast Extraction Block, with the exception of the first stage, which incorporates a convolutional stem. Within each stage, convolutional downsampling is applied at intermediate layers to facilitate the extraction of deep feature information.

This enables the extraction of features with adaptive receptive fields of varying sizes from the image. Additionally, multi-branch training enhances the model's generalization ability and robustness. During testing, the weights of each branch are consolidated into a single branch. The following formula illustrates the calculation process during training and inference:

$$X_{\text{train}} = \text{BN}_1\left(X_{0:c_p} \otimes \text{Conv}_{3\times3}\right) + \text{BN}_2\left(X_{0:c_p} \otimes \text{Conv}_{1\times1}\right) + \text{BN}_3\left(X_{0:c_p}\right) \tag{11}$$

$$X_{\text{out}} = \text{Concat}\left(X_{0:c_p}, X_{c_p:c}\right) \tag{12}$$

where $\otimes$ represents the convolution operation, c_1 represents the first c_1 channels of c, and BN represents the batch normalization. The inference phase is as follows:

$$X_{\text{out}} = \text{Concat}\left(\left(X_{0:c_p} \otimes \text{Conv}_{3\times3}\right), X_{c_p:c}\right). \tag{13}$$

The convolution kernel is fused with BN, and the three branches are merged into a single branch. We will verify the validity of our design in the experiments detailed in Section IV.

IV. Experimental Results

In accordance with the absolute scale-based object size definition established by the COCO [19] dataset for object detection, this article adopts the following categorical criteria: object instances are categorized into three groups based on their bounding box areas. Specifically, targets with pixel areas smaller than 32 × 32 pixels are classified as small-objects, and those ranging between 32 × 32 and 96 × 96 pixels are designated as medium-sized objects, while instances exceeding 96 × 96 pixels in bounding box area are identified as large objects.

This article conducts experiments on three public datasets for open-water detection: SeaDronesSee [61], aerial dataset of floating objects (AFOs) [62], and aerialbus ship (AS). We conducted experiments with various Transformer and CNN models and compared our light model, which achieved improved accuracy and speed with a lower parameter count. A comprehensive overview of the three open-access benchmark datasets employed in this study is provided in Section IV-A.

A. Datasets

1) SeaDronesSee: The SeaDronesSee dataset is designed for object detection in open-waters, particularly to support drone applications in maritime search and rescue missions. It contains images captured by drones at various altitudes (5∼260) m and viewing angles (0°∼90°), comprising 8930 training images, 1547 validation images, and 3750 test images, with resolutions ranging from 1280 × 960 to 5456 × 3632. Each image is accompanied by detailed metadata. The dataset includes six predefined fine-grained category labels (e.g., swimmers, boats, floating objects, and life-saving equipment) for manual annotation of training and validation images. Furthermore, as shown in Fig. 4, small-objects in SeaDronesSee constitute 91% of all annotations.

Fig. 5 illustrates the distribution of small-objects in terms of both count and proportion across individual images within the SeaDronesSee training set. Given the extensive scale of the dataset, sparse sampling was employed to randomly select a subset of 100 images for analysis. The horizontal axis corresponds to the image index, while the vertical axis represents the target count. In the figure, the green histogram denotes the number of small-objects within each image, whereas the grayscale histogram indicates the total number of objects present.

2) Aerial Dataset of Floating Object: The AFO dataset is the first dataset specifically designed for maritime search and rescue applications. It consists of 50 video clips (with resolutions ranging from 1280 × 720 to 3840 × 2160) captured from drone-mounted cameras, from which 3647 images were extracted. These images contain 39 991 manually annotated objects, including people and objects floating in the water.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:52:36 UTC from IEEE Xplore. Restrictions apply.

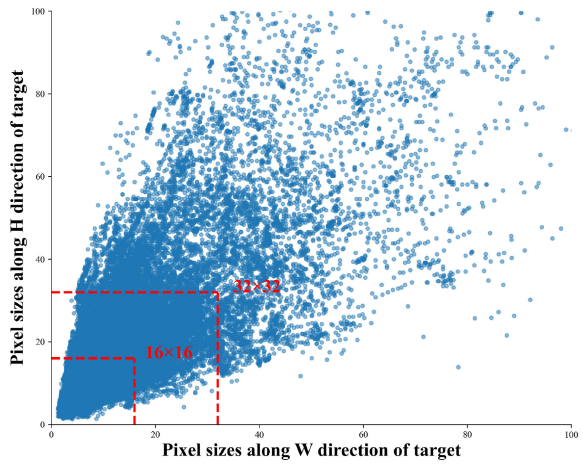


Fig. 4. Distribution of object sizes in SeaDronesSee.

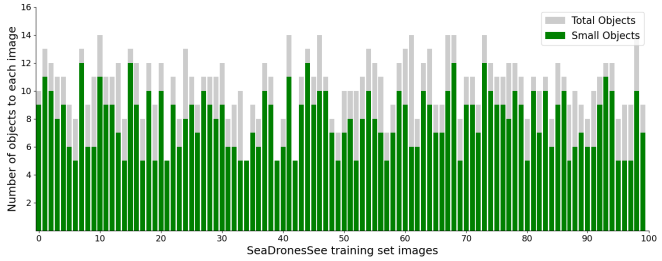


Fig. 5. Number and percentage of small-objects in the training set of SeaDronesSee.

These images are divided into three sets: training 67.4% of objects, testing 19.12% of objects, and validation 13.48% of objects.

3) Aerialbus Ship: The AS dataset is a publicly available dataset used for marine vessel detection. It primarily consists of satellite images of oceans, featuring ships of various types and sizes against different backgrounds, including ports, coastlines, and open seas. The Airbus Ship dataset includes 192 556 images at a resolution of 768 × 768, with 154 044 training images, 38 511 validation images, and 15 606 unlabeled test images. Due to the significant variation in ship size, orientation, and occlusion, as well as complex background environments, the Airbus Ship dataset requires high detection accuracy and model robustness. This makes it an important benchmark dataset for marine monitoring and ship detection.

B. Settings

In the experiment, three public datasets were used for validation: SeaDronesSee, AFO, and Airbus Ship. The model utilizes ResNet50 or ResNet101 [56], pretrained on ImageNet [70], as the backbone network to extract multiscale features. The loss function is optimized for 100 epochs using the AdamW optimizer [10], with a learning rate of 0.0001 and a weight decay set to 0.0001. Other parameters are consistent with RT-DETR. All experiments were implemented in PyTorch and conducted on an RTX 4090 GPU.

C. Evaluation Metrics

To evaluate detection quality, the COCO evaluation metric is used in this article. Precision and recall are used as performance metrics for various methods. Prediction boxes that do not contain objects are classified as false negatives (FNs). Prediction boxes containing objects with an IoU greater than 0.5 with the ground truth are classified as true positives (TPs), while those with an IoU less than 0.5 are classified as FPs. Frames per second (FPS) refers to the number of image frames processed by the model each second and is commonly used to evaluate real-time processing capability. The corresponding formula is as follows:

$$\text{Precision} = \frac{\|\text{TP}\|}{\|\text{TP}\| + \|\text{FP}\|} \tag{14}$$

$$\text{Recall} = \frac{\|\text{TP}\|}{\|\text{TP}\| + \|\text{FN}\|}. \tag{15}$$

TP, FP, and FN represent the respective counts of TPs, FPs, and FNs. Higher precision indicates greater accuracy in detecting objects, while higher recall reflects a greater proportion of objects detected. Average precision (AP) represents the average area under the precision–recall (PR) curve across all categories, while ARmax refers to the model's recall when using the maximum number of prediction boxes per image. The formula is given as follows:

$$\text{mAP} = \frac{1}{N}\sum_{n=1}^{N} \text{AP}_n \tag{16}$$

$$\text{AP} = \frac{1}{10}\sum_{i=1}^{10} \text{mAP}_{\text{IoU}=0.5+(i-1)\times 0.05} \tag{17}$$

$$\text{FPS} = \frac{N}{T}. \tag{18}$$

D. Comparative Evaluation With State of the Arts

Fig. 6 shows the detection results of OWRT-DETR, RT-DETR, and YOLO-OW on the three publicly available datasets. It can be seen that the other models have different degrees of misdetection and underdetection for small or dense targets on all three datasets, while OWRT-DETR shows its strong performance on small-objects and dense targets due to its unique design.

1) SeaDronesSee: Table I shows a comparison between other state-of-the-art models and OWRT-DETR on the SeaDronesSee dataset. The proposed model achieves the highest precision and recall with reduced FLOPs (124.8) and Parameters (38.1M). Compared to the baseline using the ResNet50 backbone, OWRT-DETR reduces FLOPs by 4.8G and parameters by 4.2M. Additionally, due to the Transformer's more powerful global modeling capability compared to CNN, OWRT-DETR achieves high precision and recall relative to the CNN-based baseline model. Furthermore, the SOEM enables OWRT-DETR to maintain strong performance in small-object detection accuracy. Compared to DINO-DETR, Focus-DETR, and H-Deformable-DETR, OWRT-DETR achieves an average improvement of 2% in AP. Compared to the baseline RT-DETR, it demonstrates an improvement of 2%–3% across various APs. Additionally, the proposed L-OWRT-DETR

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:52:36 UTC from IEEE Xplore. Restrictions apply.

TABLE I
COMPARISON OF DIFFERENT METHODS IN SEADRONESSEE

Model	Backbone	#Epochs	#Params (M)	GFLOPs	$FPS_{bs=1}$	AP^{val} (%)	AP^{val}_{50} (%)	AP^{val}_{75} (%)	AP^{val}_{S}
CNN Based									
Faster-RCNN [17]	R50	24	43.1	223.6	-	-	59.79	-	-
SSD [18]	R50	24	24.9	88.1	-	-	58.93	-	-
YOLOv8 [63]	-	200	43.7	165.7	52.3	39.3	65.5	40.8	28.3
YOLOv9 [64]	-	200	25.3	102.3	100.1	39.3	65.4	40.0	27.0
YOLOv10 [65]	-	200	24.3	120.0	132.5	37.2	64.1	38.4	23.1
YOLOv11-L [66]	-	200	25.2	86.6	**143.7**	38.4	64.3	39.5	26.2
YOLOv11-X [66]	-	200	56.8	195.5	87.8	39.0	65.5	39.6	25.6
YOLO-OW [33]	-	200	42.1	94.8	61.3	39.8	64.6	35.7	24.7
Transformer Based									
Anchor-DETR [27]	R50	50	36.8	172.4	-	39.4	74.9	38.5	31.5
Anchor-DETR [27]	R101	50	55.8	-	-	40	76.7	38.1	27.9
Conditional-DETR [67]	R50	50	43.2	63.6	-	29.7	62.5	24.3	14.3
Conditional-DETR [67]	R101	50	62.1	226.1	-	32.1	63.0	28.9	19.2
Deformable-DETR [26]	R50	50	39.8	170.3	-	38.2	72.2	35.3	25.5
Deformable-DETR [26]	R101	50	58.8	226.1	-	39.6	76.4	36.3	31.7
DAB-DETR [29]	R50	50	47.2	163.5	-	49.2	83.5	50.2	42.7
DAB-DETR [29]	R101	50	66.1	226.1	-	50.3	81.9	52.8	43.6
DN-DETR [28]	R50	50	47.3	166.9	-	49.5	82.7	50.9	42.7
DN-DETR [28]	R101	50	66.2	226.1	-	49.4	80.5	**53.1**	42.2
H-Deformable-DETR [68]	R50	50	46.2	168.9	-	39.5	73.1	38.1	30.9
Focus-DETR [69]	R50	36	49.3	113.7	-	49.8	82.9	51.4	43.5
Focus-DETR [69]	R101	36	68.9	226.1	-	50.1	83.1	52.2	44.6
DINO-4scale [34]	R50	12	46.6	178.5	-	48.0	81.9	48.4	40.0
DINO-4scale [34]	R101	12	65.6	226.1	-	48.5	82.5	48.8	44.9
RT-DETR [23]	R50	72	42.3	129.6	61.2	48.2	82.3	48.3	42.8
RT-DETR [23]	R101	72	74.6	247.1	55.3	49.8	83.7	51.3	44.1
OWRT-DETR-R50 (ours)	R50	100	38.1	124.8	59.4	50.8	**85.3**	50.4	44.3
OWRT-DETR-R101 (ours)	R101	100	63.8	219.2	51.2	**51.5**	**86.3**	51.8	**45.2**
L-OWRT-DETR (ours)	LFENet	100	**19.1**	**54.4**	72.4	50.3	84.9	47.9	44.1

TABLE II
COMPARISON OF DIFFERENT METHODS IN AFO

Model	Backbone	#Epochs	AP^{val} (%)	AP^{val}_{50} (%)	AP^{val}_{S}
YOLOv10 [65]	-	200	30.5	62.7	7.2
YOLOv11 [66]	-	200	31.2	63.3	7.0
YOLO-OW [33]	-	200	34.6	67.3	9.1
DAB-DETR [29]	R50	50	44.7	81.2	9.7
DN-DETR [28]	R50	50	42.9	75.9	8.7
H-Deformable-DETR [68]	R50	50	38.4	69.6	7.2
Focus-DETR [69]	R50	36	44.8	80.2	10.2
DINO-4scale [34]	R50	12	43.3	76.2	8.4
RT-DETR [23]	R50	72	43.8	79.2	9.4
OWRT-DETR-R50 (ours)	R50	100	**45.9**	**82.3**	**11.4**
L-OWRT-DETR (ours)	LFENet	100	43.6	78.9	9.7

achieves better results in both AP and FPS with FLOPs and parameters under half of those in the baseline model.

2) Aerial Dataset of Floating Object: To verify the generalization ability of the proposed method on open-water UAV images, OWRT-DETR was tested on the AFO dataset. As shown in Table II, OWRT-DETR demonstrates significant advantages over other models, with improvements of 0.9% and 1.1% in AP and AP_{50}, respectively, over state-of-the-art (SOTA) methods. Notably, for small-objects, AP_{Small} is increased by 1.2%. This further demonstrates OWRT-DETR excellent performance in detecting small-object detection.

3) Aerialbus Ship: As shown in Table III, to further evaluate the model's generalization ability, we tested it on remote sensing images with low resolution and poor quality. Compared to the SOTA model, AP is improved by 5.1%, AP_{50} by 4.2%, and AP_S by 5.9%. In summary, OWRT-DETR not only improves performance on UAV aerial data, but also demonstrates significant effectiveness on low-quality remote sensing datasets. Additionally, the performance of our proposed L-OWRT-DETR significantly surpasses that of the SOTA model.

E. Grad-CAM Map Analysis

Gradient-weighted class activation mapping (Grad-CAM) [71], a visualization technique, utilizes gradients to estimate the spatial significance of positions in an image. As shown in Fig. 7, Grad-CAM visualizations are generated for YOLO11, YOLO-OW, the baseline model, and OWRT-DETR after cross-scale interactions at stages S'_3, S'_4, and S'_5. As illustrated, features from S'_3, S'_4, and S'_5 enable a qualitative analysis of each model's adaptability to spatial scale. For images from the SeaDronesSee dataset, Grad-CAM visualizations focus on two

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:52:36 UTC from IEEE Xplore. Restrictions apply.

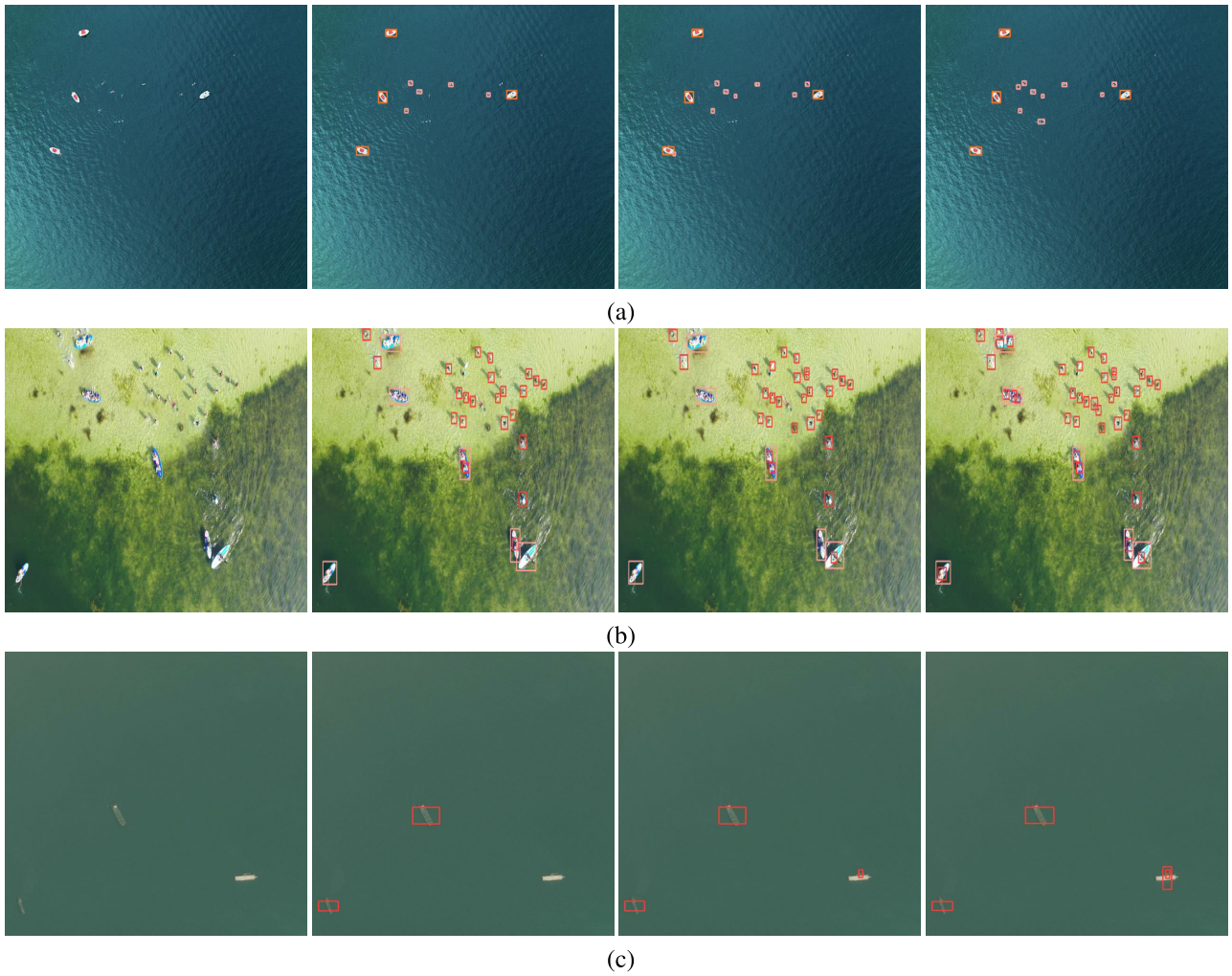


Fig. 6. Visualization of detection results among YOLO-OW, baseline, OWRT-DETR in SeaDronesSee, AFO, and AS datasets. The first column is the original images. (a) Results in SeaDronesSee dataset. (b) Results in AFO dataset. (c) Results in AS dataset.

TABLE III
COMPARISON OF DIFFERENT METHODS IN AIRBUS SHIP

Model	Backbone	#Epochs	AP^{val} (%)	AP^{val}_{50} (%)	AP^{val}_{S}
YOLOv10 [65]	-	200	40.2	79.3	30.4
YOLOv11 [66]	-	200	39.2	76.5	26.2
DAB-DETR [29]	R50	50	43.9	79.1	32.7
DN-DETR [28]	R50	50	42.8	80.0	31.5
H-Deformable-DETR [68]	R50	50	39.4	76.6	28.3
Focus-DETR [69]	R50	36	44.8	80.2	33.4
DINO-4scale [34]	R50	12	43.3	76.2	32.4
RT-DETR [23]	R50	72	45.5	78.6	34.6
OWRT-DETR-R50 (ours)	R50	100	**50.6**	**82.8**	**40.5**
L-OWRT-DETR (ours)	LFENet	100	**46.1**	**80.4**	**35.9**

categories, "boat" and "swimmer." Given the considerable distance from which these images are captured, the object scale is significantly reduced, presenting a substantial challenge to the models.

In Fig. 7(a) and (b), the CNN-based YOLO model demonstrates strong attention to "boat" objects though it exhibits limited focus on the smaller "swimmer" category and is affected by water surface glare, resulting in potential misclassifications. In Fig. 7(c), the baseline model effectively mitigates glare but still encounters interference from water surface ripples and lacks sufficient feature representation for smaller objects. As shown in Fig. 7(d), OWRT-DETR enhances attention to smaller objects through its SOEM and enriches each level with detailed small-object information via cross-scale interaction in the CFPIM. This approach strengthens global object attention, enhancing detection accuracy and reducing FPs.

F. Ablation Studies

To analyze the importance of each component in OWRT-DETR, we progressively applied the CFPIM, SOEM, and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:52:36 UTC from IEEE Xplore. Restrictions apply.

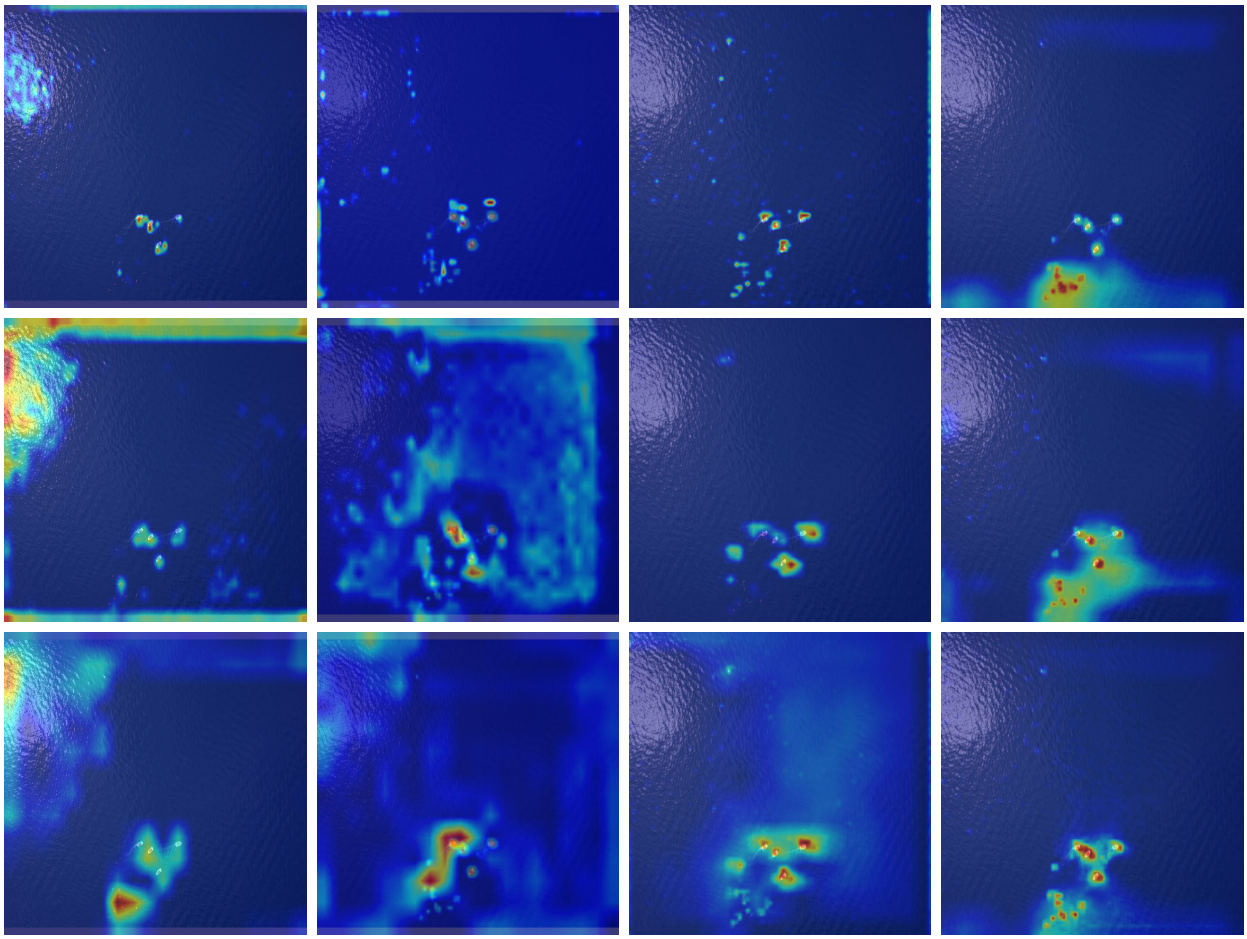


Fig. 7. Grad-CAM visualization results of (a) Yolo11, (b) YoloOW, (c) RT-DETR, and (d) OWRT-DETR models on $S3 - S5$ after cross-scale feature interaction.

TABLE IV
ABLATION EXPERIMENTS FOR CFPIM, SOEM, AND MSSF IN SEADRONESSEA

CFPIM	SOEM	MSSF	AP^{val} (%)	AP^{val}_{50} (%)	AP^{val}_{S}	#Params (M)	GFLOPs
-	-	-	48.2	82.3	42.8	42.3	134.8
✓	-	-	50.0	82.9	43.1	43.2	136.9
-	✓	-	49.8	83.6	43.8	44.3	146.9
-	-	✓	49.3	82.5	41.9	33.2	98.4
✓	✓	-	49.7	83.8	43.7	44.4	156.5
✓	-	✓	50.3	83.1	42.5	38.9	112.1
-	✓	✓	50.5	84.3	44.2	37.5	120.5
✓	✓	✓	50.8	85.3	44.3	38.1	125.1

MSSF in the baseline to verify their effectiveness. The ablation experiments were conducted in the SeaDronesSee dataset. Results are shown in Table IV, where "✓" indicates that the module is used.

1) CFPIM: As shown in Table IV, the integration of CFPIM leads to a significant enhancement across all evaluation metrics, particularly in terms of average accuracy. The AP increases from 48.2 to 50.0, while the small-object precision AP_{Small} improves from 42.8 to 43.1. These results validate that the multiscale interactions facilitated by CFPIM effectively enable the model to learn diverse scale-specific features, thereby enhancing its overall detection performance.

2) Small-Object Enhancement Module : As shown in Table IV, the incorporation of SOEM enhances all evaluation metrics, particularly the AP for small-objects (AP_{Small}), which increases from 42.8 to 43.8. Furthermore, we conducted an analysis of the feature maps before and after processing by SOEM, as illustrated in Fig. 8. The highlighted regions in the feature maps indicate the model's areas of focus. It is evident that the feature maps prior to SOEM processing struggle to adequately concentrate on small-object information and are often influenced by background noise, such as wave ripples. After SOEM fusion, the module effectively suppresses complex background interference while significantly improving

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:52:36 UTC from IEEE Xplore. Restrictions apply.

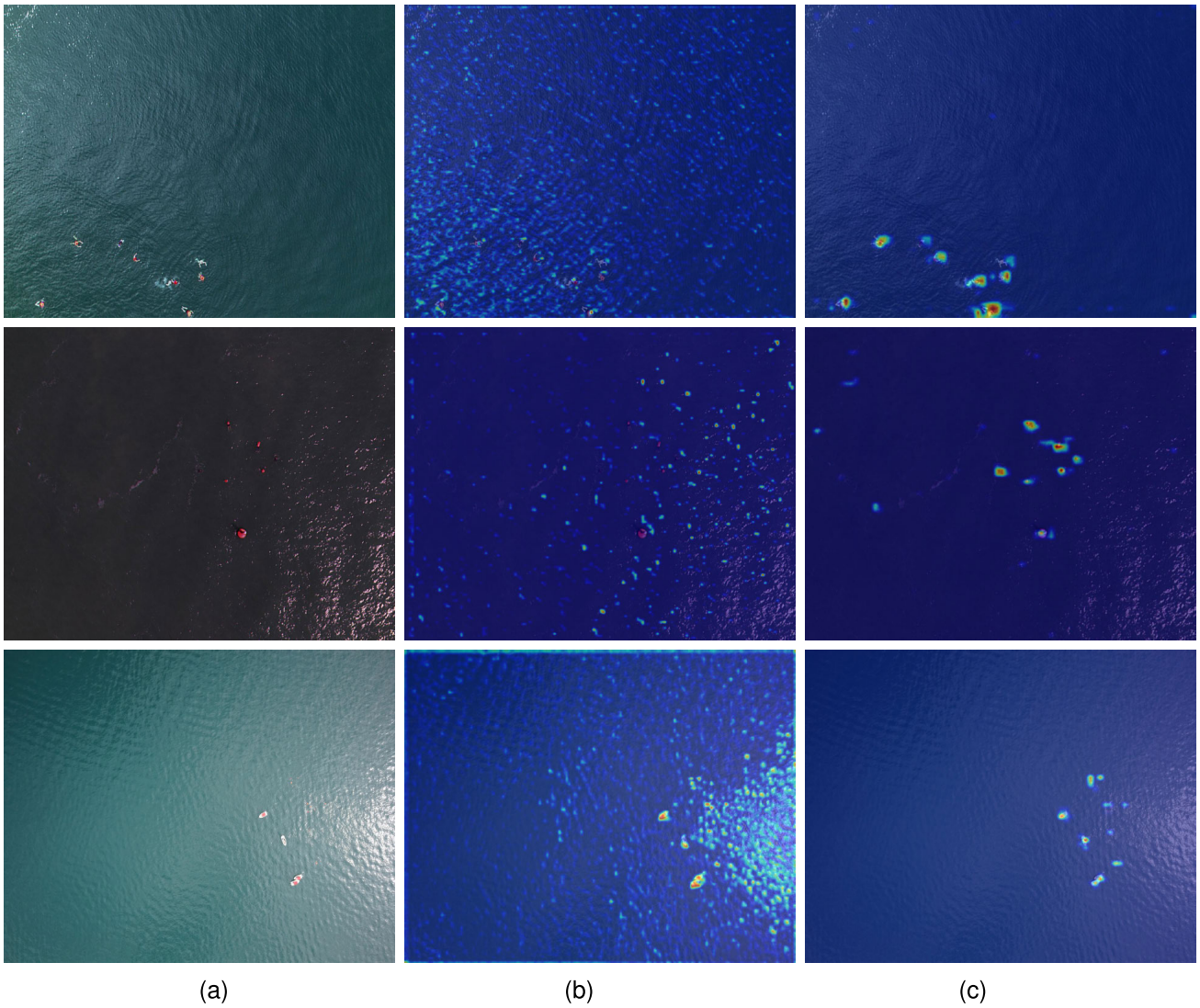


Fig. 8. Influence of SOEM on feature extraction in SeaDronesSee dataset. (a) Three distinct original images from the sample range. (b) Grad-CAM visualization of the three original images prior to SOEM processing. (c) Grad-CAM visualization of the three images after SOEM processing. The brighter color represents that the model pays more attention to that area.

TABLE V
ABLATION EXPERIMENTS FOR CSF BLOCK IN SEADRONESSEE

Method	AP	AP_{50}	Params	GFLOPs
RTDETR(Rep=3)	48.2	82.3	42.3	134.8
CSF(Rep=1)	49.3	82.5	33.2	98.4
CSF(Rep=2)	49.5	82.8	35.4	108.3
CSF(Rep=3)	49.9	83.1	37.6	119.0
CSF(Rep=4)	50.2	83.4	39.7	130.6
CSF(Rep=5)	49.6	82.7	41.9	140.4

the model's ability to perceive and emphasize small-object features.

3) Multiscale Sensing Fusion: As shown in Table IV, the incorporation of MSSF leads to significant improvements across all performance metrics. Additionally, Table V analyzes the impact of varying repetition counts (Rep) in the cross-sensing fusion (CSF) block of the MSSF module on model performance and parameter efficiency. The experimental results demonstrate that increasing Rep improves AP, peaking at 49.9 when Rep $=$ 4. However, to balance computational overhead and real-time deployment requirements, Rep $=$ 1 was ultimately selected as the optimal hyperparameter. Compared to RTDETR, the proposed OWRT-DETR with MASA attention decoupling reduces model parameters and computational costs by 25%, while simultaneously achieving significant performance gains.

Through the above analysis of ablation experiments, it can be concluded that the proposed modules CFPIM, SOEM, and MSSF all steadily improve the performance of OWRT-DETR.

G. Lightweight Comparison Experiment

To verify the lightweight effect of L-OWRT-DETR, Block, FasterBlock, and LFEBlock in ResNet are compared and the results are shown in Table VI. It can be seen that although ResNet has higher accuracy compared to FasterBlock, the number of parameters and memory access requirements are larger, having more computational redundancy and memory access. After we have performed multibranching and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:52:36 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
LIGHTWEIGHT EXPERIMENT FOR L-OWRT-DETR

Method	AP	Params	GFLOPs	FPS
ResBlock(R18)	47.8	20.2	58.6	65.5
FasterBlock(ratio=0.25)	47.2	17.1	51.2	75.6
LFEBlock(ratio=1)	50.1	22.7	64.0	69.2
LFEBlock(ratio=0.5)	49.7	18.0	52.7	71.5
LFEBlock(ratio=0.25)	49.5	16.8	49.8	74.1

reparameterized fusion branching, LFEBlock not only has faster speed, but it can optimize the speed, accuracy, and memory requirements more effectively. In addition, in order to obtain the optimized LFEBlock with different channel scaling ratios, we did comparison experiments with different ratios of LFEBlock, and it can be seen that although the LFEBlock with a ratio of 0.25 reduces the accuracy by 0.6%, the advantageous performance gained in speed and memory access far outweighs the shortcomings that the lost accuracy comes to bring.

V. CONCLUSION

This article proposed a DETR variant, OWRT-DETR, which adapts to different spatial scales to improve the detection performance of objects in open-waters, especially small-scale objects, in drone aerial images. In particular, three new modules SOEM, CFPIM, and MSSF are introduced. First, SOEM extracts multiscale sequence features through 3-D convolution, followed by spatial and channel attentions, which helps to preserve the features of small-objects and address foreground–background confusion. Second, MSSF adapts to different feature scales while modeling multiscale global relationships without increasing the number of parameters, effectively enhancing the model's feature representation capability. Additionally, CFPIM is introduced to guide cross-scale learning, establishing contextual information interactions across different scales. Moreover, the simplified version of OWRT-DETR, L-OWRT-DETR, reconstructs the backbone using PConv and Rep structures. Compared to OWRT-DETR, L-OWRT-DETR is faster, has fewer parameters, and requires less computation, with minimal accuracy loss. Experiments demonstrate that it outperforms the SOTA models on two marine detection datasets, SeaDronesSee and AFO. We further tested it on the Airbus Ship remote sensing dataset, which presents challenges including low resolution, poor quality, unclear features, complex degradation, extensive water surface, and sparse objects, making feature extraction difficult. AP results significantly exceeded those of other benchmark models on this dataset. While OWRT-DETR has achieved promising results in open-water small-object detection and shows the future potential for airborne applications, it still has some limitations.

1) Hardware deployment requires further optimization of speed and memory utilization.
2) Presently, the model has only been validated on datasets without environmental interference. For open-water images in complex environments, background interference is typically strong, target features are less distinct, and weather conditions are often challenging.

Moreover, investigations reveal that existing networks typically encounter bottlenecks when detecting marine objects with single-modality input. Multimodal combinations can provide the model with more effective feature representations. As a result, multimodal detection may represent the future direction for small-object detection.

REFERENCES

[1] H. Wang, S. Tong, A. Wang, W. Zhang, Z. Hu, and Z. Peng, "Heterogeneous cross domain coordinated control of ASV-AUV system for maritime search and rescue," *Ocean Eng.*, vol. 306, Aug. 2024, Art. no. 117950.

[2] J. Wu, L. Cheng, S. Chu, and Y. Song, "An autonomous coverage path planning algorithm for maritime search and rescue of persons-in-water based on deep reinforcement learning," *Ocean Eng.*, vol. 291, Jan. 2024, Art. no. 116403.

[3] S. W. Cho, H. J. Park, H. Lee, D. H. Shim, and S.-Y. Kim, "Coverage path planning for multiple unmanned aerial vehicles in maritime search and rescue operations," *Comput. Ind. Eng.*, vol. 161, Nov. 2021, Art. no. 107612.

[4] C. Deiana, V. Maheshri, and G. Mastrobuoni, "Migrants at sea: Unintended consequences of search and rescue operations," *Amer. Econ. J., Econ. Policy*, vol. 16, no. 2, pp. 335–365, May 2024.

[5] T. Yang, Z. Jiang, R. Sun, N. Cheng, and H. Feng, "Maritime search and rescue based on group mobile computing for unmanned aerial vehicles and unmanned surface vehicles," *IEEE Trans. Ind. Informat.*, vol. 16, no. 12, pp. 7700–7708, Dec. 2020.

[6] S. Lee and J. R. Morrison, "Decision support scheduling for maritime search and rescue planning with a system of UAVs and fuel service stations," in *Proc. Int. Conf. Unmanned Aircr. Syst. (ICUAS)*, Jun. 2015, pp. 1168–1177.

[7] T. Laperrière-Robillard, M. Morin, and I. Abi-Zeid, "Supervised learning for maritime search operations: An artificial intelligence approach to search efficiency evaluation," *Expert Syst. Appl.*, vol. 206, Nov. 2022, Art. no. 117857, doi: 10.1016/j.eswa.2022.117857.

[8] S. Qi, B. Lin, Y. Deng, X. Chen, and Y. Fang, "Minimizing maximum latency of task offloading for multi-UAV-assisted maritime search and rescue," *IEEE Trans. Veh. Technol.*, vol. 73, no. 9, pp. 13625–13638, Sep. 2024.

[9] X. Liang, J. Zhang, L. Zhuo, Y. Li, and Q. Tian, "Small object detection in unmanned aerial vehicle images using feature fusion and scaling-based single shot detector with spatial context analysis," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 30, no. 6, pp. 1758–1770, Jun. 2020.

[10] H. Ruan, W. Qian, Z. Zheng, and Y. Peng, "A decoupled semantic–detail learning network for remote sensing object detection in complex backgrounds," *Electronics*, vol. 12, no. 14, p. 3201, Jul. 2023.

[11] Y. Zhang, M. Ye, G. Zhu, Y. Liu, P. Guo, and J. Yan, "FFCA-YOLO for small object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, doi: 10.1109/TGRS.2024.3363057.

[12] M. Hearst, S. Dumais, E. Osuna, J. Platt, and B. Scholkopf, "Support vector machines," *IEEE Intell. Syst. Appl.*, vol. 13, no. 4, pp. 18–28, Aug. 1998.

[13] N. Dalal and B. Triggs, "Histograms of oriented gradients for human detection," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, San Diego, CA, USA, Jun. 2005, pp. 886–893.

[14] F. Han, Y. Shan, R. Cekander, H. S. Sawhney, and R. Kumar, "A two-stage approach to people and vehicle detection with HOG-based SVM," in *Proc. Performance Metrics Intelligent Syst. Workshop*, 2006, pp. 133–140.

[15] M. Bilal and M. S. Hanif, "Benchmark revision for HOG-SVM pedestrian detector through reinvigorated training and evaluation methodologies," *IEEE Trans. Intell. Transp. Syst.*, vol. 21, no. 3, pp. 1277–1287, Mar. 2020.

[16] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 779–788.

[17] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2016.

[18] W. Liu et al., "SSD: Single shot multibox detector," in *Proc. 14th Eur. Conf. Comput. Vis. (ECCV)*, Cham, Switzerland. Amsterdam, The Netherlands: Springer, Oct. 2016, pp. 21–37.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:52:36 UTC from IEEE Xplore. Restrictions apply.

[19] T. Lin et al., "Microsoft COCO: Common objects in context," in *Proc. Eur. Conf. Comput. Vis.*, 2014, pp. 740–755.

[20] M. Everingham, L. Van Gool, C. K. I. Williams, J. Winn, and A. Zisserman, "The Pascal visual object classes (VOC) challenge," *Int. J. Comput. Vis.*, vol. 88, no. 2, pp. 303–338, 2010.

[21] Z. Cai and N. Vasconcelos, "Cascade R-CNN: Delving into high quality object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6154–6162.

[22] Z. Ge, S. Liu, F. Wang, Z. Li, and J. Sun, "YOLOX: Exceeding YOLO series in 2021," 2021, *arXiv:2107.08430*.

[23] Y. Zhao et al., "DETRs beat YOLOs on real-time object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 16965–16974.

[24] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, Jun. 2017, pp. 5998–6008.

[25] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2020, pp. 213–229.

[26] X. Zhu, W. Su, L. Lu, B. Li, X. Wang, and J. Dai, "Deformable DETR: Deformable transformers for end-to-end object detection," 2020, *arXiv:2010.04159*.

[27] Y. Wang, X. Zhang, T. Yang, and J. Sun, "Anchor DETR: Query design for transformer-based detector," in *Proc. AAAI Conf. Artif. Intell.*, vol. 36, no. 3, Jun. 2022, pp. 2567–2575.

[28] F. Li, H. Zhang, S. Liu, J. Guo, L. M. Ni, and L. Zhang, "DN-DETR: Accelerate DETR training by introducing query denoising," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 13619–13627.

[29] S. Liu et al., "DAB-DETR: Dynamic anchor boxes are better queries for DETR," 2022, *arXiv:2201.12329*.

[30] D. Xu and Y. Wu, "FE-YOLO: A feature enhancement network for remote sensing target detection," *Remote Sens.*, vol. 13, no. 7, p. 1311, Mar. 2021.

[31] L. Shen, B. Lang, and Z. Song, "CA-YOLO: Model optimization for remote sensing image object detection," *IEEE Access*, vol. 11, pp. 64769–64781, 2023.

[32] H. Zhao, H. Zhang, and Y. Zhao, "YOLOv7-sea: Object detection of maritime UAV images based on improved YOLOv7," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. Workshops (WACVW)*, Jan. 2023, pp. 233–238.

[33] J. Xu et al., "YoloOW: A spatial scale adaptive real-time object detection neural network for open water search and rescue from UAV aerial imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, doi: 10.1109/TGRS.2024.3395483.

[34] H. Zhang et al., "DINO: DETR with improved DeNoising anchor boxes for end-to-end object detection," 2022, *arXiv:2203.03605*.

[35] Z. Yao, J. Ai, B. Li, and C. Zhang, "Efficient DETR: Improving end-to-end object detector with dense prior," 2021, *arXiv:2104.01318*.

[36] Q. Chen et al., "Group DETR: Fast DETR training with group-wise one-to-many assignment," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 6610–6619.

[37] Y. Pu et al., "Rank-DETR for high quality object detection," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2023.

[38] T. Y. Lin, P. Dollàr, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 2117–2125.

[39] K. Wang, J. H. Liew, Y. Zou, D. Zhou, and J. Feng, "PANet: Few-shot image semantic segmentation with prototype alignment," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9197–9206.

[40] G. Ghiasi, T.-Y. Lin, and Q. V. Le, "NAS-FPN: Learning scalable feature pyramid architecture for object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 7036–7045.

[41] M. Tan, R. Pang, and Q. V. Le, "EfficientDet: Scalable and efficient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10781–10790.

[42] S. Liu, D. Huang, and Y. Wang, "Learning spatial fusion for single-shot object detection," 2019, *arXiv:1911.09516*.

[43] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.

[44] H. Wang et al., "Spatial attention for multi-scale feature refinement for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshop (ICCVW)*, Oct. 2019, pp. 64–72.

[45] G. Cheng, C. Lang, M. Wu, X. Xie, X. Yao, and J. Han, "Feature enhancement network for object detection in optical remote sensing images," *J. Remote Sens.*, vol. 2021, Jan. 2021.

[46] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, "Rethinking the inception architecture for computer vision," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2818–2826.

[47] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2261–2269.

[48] K. He, X. Zhang, S. Ren, and J. Sun, "Identity mappings in deep residual networks," 2016, *arXiv:1603.05027*.

[49] C. Szegedy et al., "Going deeper with convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2015, pp. 1–9.

[50] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," *Commun. ACM*, vol. 60, no. 6, pp. 84–90, May 2017, doi: 10.1145/3065386.

[51] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.

[52] Q. Fan, H. Huang, M. Chen, H. Liu, and R. He, "RMT: Retentive networks meet vision transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 5641–5651.

[53] Y. Sun et al., "Retentive network: A successor to transformer for large language models," 2023, *arXiv:2307.08621*.

[54] X. Li, W. Wang, X. Hu, and J. Yang, "Selective kernel networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 510–519.

[55] Y. Li et al., "LSKNet: A foundation lightweight backbone for remote sensing," *Int. J. Comput. Vis.*, vol. 133, no. 3, pp. 1410–1431, Mar. 2025.

[56] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[57] X. Zhang, X. Zhou, M. Lin, and J. Sun, "ShuffleNet: An extremely efficient convolutional neural network for mobile devices," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6848–6856.

[58] K. Han, Y. Wang, Q. Tian, J. Guo, C. Xu, and C. Xu, "Ghostnet: More features from cheap operations," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2020, pp. 1580–1589.

[59] J. Chen et al., "Run, don't walk: Chasing higher FLOPS for faster neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Sep. 2023, pp. 12021–12031.

[60] X. Ding, X. Zhang, N. Ma, J. Han, G. Ding, and J. Sun, "RepVGG: Making VGG-style ConvNets great again," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 13733–13742.

[61] L. A. Varga, B. Kiefer, M. Messmer, and A. Zell, "SeaDronesSee: A maritime benchmark for detecting humans in open water," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2022, pp. 3686–3696.

[62] J. Gąsienica-Józkowy, M. Knapik, and B. Cyganek, "An ensemble deep learning method with optimized weights for drone-based water rescue and surveillance," *Integr. Comput.-Aided Eng.*, vol. 28, no. 3, pp. 221–235, Jun. 2021.

[63] R. Varghese and S. M., "YOLOv8: A novel object detection algorithm with enhanced performance and robustness," in *Proc. Int. Conf. Adv. Data Eng. Intell. Comput. Syst. (ADICS)*, Apr. 2024, pp. 1–6.

[64] C.-Y. Wang, I.-H. Yeh, and H.-Y. M. Liao, "YOLOv9: Learning what you want to learn using programmable gradient information," 2024, *arXiv:2402.13616*.

[65] A. Wang et al., "YOLOv10: Real-time end-to-end object detection," 2024, *arXiv:2405.14458*.

[66] G. Jocher and J. Qiu. (2024). *Ultralytics YOLO11*. [Online]. Available: https://github.com/ultralytics/ultralytics

[67] D. Meng et al., "Conditional DETR for fast training convergence," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 3651–3660.

[68] D. Jia et al., "DETRs with hybrid matching," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 19702–19712.

[69] D. Zheng, W. Dong, H. Hu, X. Chen, and Y. Wang, "Less is more: Focus attention for efficient DETR," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 6674–6683.

[70] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Miami, FL, USA, Jun. 2009, pp. 248–255.

[71] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 618–626.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:52:36 UTC from IEEE Xplore. Restrictions apply.